

Curriculum Vitae

AREAS OF SPECIALIZATION	Software Engineering, Machine Learning, Human Language Technology, Speech Recognition, Language Modeling, Signal Processing, Pattern Recognition, Statistical Modeling
Contact	Email: amir.harati@gmail.com
PERSONAL	First Name: AmirHossein Last Name: HaratiNejadTorbati Nick Name: Amir Harati Legal Status: Canadian citizen, US Green Card holder github: https://github.com/amirharati linkedin: https://www.linkedin.com/in/amir-harati-44478045/
EDUCATION	Ph.D. in Computer and Electrical Engineering (January 2010 – July 2015): Department of Electrical and Computer Engineering, Temple University, 1947 North 12th Street, Philadelphia, Pennsylvania, 19122, GPA: 4.0/4.0. PhD Dissertation title: <i>Nonparametric Bayesian Approaches for Acoustic Modeling</i>. Adviser: Dr. Picone. M.S. in Electrical Engineering (September 2005 – June 2007): Department of Electrical and Computer Engineering, K.N. Toosi University of Technology, Seyedkhandan, and Dr. Shariati Ave Tehran, Iran. Major: Communication System, GPA: 18.17/20. Thesis title: Simulation and DSP Implementation of an LDPC Codec to determine the effective parameters for its performance. Adviser: Dr. Ahmadian. B.S. in Electrical Engineering (September 2000 – December 2004): Department of Electrical and Computer Engineering, Tabriz University, 29 Bahman Blvd., Tabriz, Iran. Major: Communication Engineering, High School Diploma (September 1997 – May 2000): Hasheminejad (NODET) High School, Ferdowsi Square, Mashhad, Iran. Emphasis: Mathematics and Physics, (NODET schools are designed for talented students and use a very competitive entrance exam as an entrance requirement.)
SKILLS	• C/C++, Python, gcc/g++, Unix, bash, HPC, Matlab, • Pytorch, Tensorflow, Keras, sklearn, nltk, rasa, spacy, flask • Sagemaker, Lambda, Cloud computing (mostly AWS ecosystem) • Machine Learning, Deep Learning, Acoustic modeling, Language Modeling, Signal Processing, Speech Processing, Statistical Modeling, Nonparametric Bayesian Modeling, Signal processing for EEG, EEG pattern recognition. • Speech recognition tools including HTK, ISIP and Kaldi. Familiar with (used them at some point): Mapreduce/spark, Elasticsearch, Mongodb, Scala, GRPC, SIMULINK, openMP, Spice, Multisim, Perl, PHP, MySQL, AJAX, PYQT, Assembly, Visual C++ (MFC), Visual Basic, HTML, Code Composer, TMS320C5510, Verilog, Valgrind, Dart, Flutter, nextjs, typescript, javascript.
PATENTS	1. A. Harati Nejad Torbati, et al. (2019), "System And Method Of Automatic Interpretation Of EEG Signals Using A Deep Learning Statistical Mode". U.S. Patent Application No. 15/560,658.

2. A. Harati Nejad Torbati et al. (2020), "Systems and methods for mental health assessment", U.S. Patent No. 10,748,644. Washington, DC: U.S. Patent and Trademark Office.
3. Elizabeth E Shriberg, Yang Lu, Amir Harati (2020), "Confidence evaluation to measure trust in behavioral health survey results", U.S. Patent Application No. 16/377,090.
4. Hartai et al. (2022), "Acoustic and natural language processing models for speech-based screening and monitoring of behavioral health conditions", U.S. Patent Application No. 17726999.
5. Tomasz Rutowski, A. Harati Nejad Torbati, Elizabeth E Shriberg (2024), "Systems and methods for multi-language adaptive mental health risk assessment from spoken and written language", US Patent Application No. 18263650.

PUBLICATIONS
Refreed Journals

1. Lin, D., Nazreen, T., Rutowski, T., Lu, Y., Harati, A., Shriberg, E., ... & Aratow, M. (2022). Feasibility of a machine learning-based smartphone application in detecting depression and anxiety in a generally senior population. *Frontiers in Psychology*, 13.
2. Meysam Golmohammadi, Amir Hossein Harati Nejad Torbati, Silvia Lopez de Diego, Iyad Obeid, Joseph Picone (2017). Automatic Analysis of EEGs Using Big Data and Hybrid Deep Learning Architectures. *Transactions on Frontiers in Human Neuroscience*, 13(76).
3. Harati, A., Picone J. (2016), A Doubly Hierarchical Dirichlet Process Hidden Markov Model with a Non-Ergodic Structure. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 24(1), 174-184.
4. Sahu, A., Saleheen, F., Oleksyuk, V., McGoverin, Cushla Pleshko, N., Harati, A., Picone, J., & Won, C.-H. (2014). Characterization of Mammary Tumors Using Noninvasive Tactile and Hyperspectral Imaging. *IEEE Sensors Journal*, 14(10), 3337 - 3344
5. Harati, A., & Picone, J. (2014). Predicting Search Term Reliability for Spoken Term Detection Systems. *International Journal of Speech Technology*, 17(1), 1-9.
6. Parker, D., Picone, J., Harati, A., Lu, S., Jenkyns, M., & Polgreen, P. (2013). Detecting paroxysmal coughing from pertussis cases using voice recognition technology. *PLoS ONE*, 8(12), e82971.

PUBLICATIONS:
Conference
Proceedings

7. R. Róžański, Elizabeth Shriberg; Yang Lu; Amir Harati; Tomasz Rutowski; Piotr Chlebek, Tulio Goulart, Ricardo Oliveira (2023), "Probabilistic Performance Bounds for Evaluating Depression Models Given Noisy Self-Report Labels," 2023 IEEE International Conference on Bioinformatics and Biomedicine (BIBM), Istanbul, Turkiye, 2023, pp. 4827-4833, doi: 10.1109/BIBM58861.2023.10385617.
8. Amir Harati, Elizabeth Shriberg, Tomasz Rutowski, Piotr Chlebek, Yang Lu, Ricardo Oliveira (2021), Speech-Based Depression Prediction Using Encoder-Weight-Only Transfer Learning and a Large Corpus. *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. Toronto, Canada (online).
9. Tomek Rutowski, Elizabeth Shriberg, Amir Harati, Yang Lu, Ricardo Oliveira, Piotr Chlebek (2021). Cross-Demographic Portability of Deep NLP-Based Depression Models. *Proceedings of IEEE Spoken Language Technology Workshop (SLT)*. Shenzhen, China (online).

10. Y Lu, A Harati, T Rutowski, R Oliveira, P Chlebek, E Shriberg (2020). Robust Speech and Natural Language Processing Models for Depression Screening. *Proceedings of IEEE Signal Processing in Medicine and Biology Symposium (SPMB)*. Philadelphia, USA (online).
11. Tomasz Rutowski, Elizabeth Shriberg, Amir Harati, Yang Lu, Piotr Chlebek, Ricardo Oliveira (2020). Depression and Anxiety Prediction Using Deep Language Models and Transfer Learning. *Proceedings of 7th International Conference on Behavioural and Social Computing (BESC)*. Bournemouth, UK (online).
12. Piotr Chlebek, Elizabeth Shriberg, Yang Lu, Tomasz Rutowski, Amir Harati, Ricardo Oliveira (2020). Comparing speech recognition services for HCI applications in behavioral health. *Proceedings of UbiComp '21*. Virtual, USA.
13. Rutowski T., Harati A., Lu Y., Shriberg E. (2019), Optimizing Speech-Input Length for Speaker-Independent Depression Classification, *Proceeding of InterSpeech*, Graz, Austria
14. Harati A., Picone J. (2016), A Nonparametric Bayesian Approach for Spoken Term detection by Example Query, *Proceeding of InterSpeech*, San Francisco , USA
15. Harati A., Picone J. (2016), A Nonparametric Bayesian Approach for Automatic Discovery of a Lexicon and Acoustic Units, *Proceeding of SLT workshop*, San Diego, USA
16. Harati, A., et al. (2015). Improved EEG Event Classification Using Differential Energy. In *Proceedings of the IEEE Signal Processing in Medicine and Biology Symposium* (pp. 1–4). Philadelphia, Pennsylvania, USA.
17. Harati, A., et al. (2014). THE TUH EEG CORPUS: A Big Data Resource for Automated EEG Interpretation. *Proceedings of the IEEE Signal Processing in Medicine and Biology Symposium* (in press). Philadelphia, Pennsylvania, USA.
18. Harati, A., Picone, J., & Sobel, M. (2014). A Left-to-Right HDP-HMM with HDPM Emissions. *Proceedings of the Conference on Information Sciences and Systems* (pp. 1–6). Princeton, New Jersey, USA.
19. Steinberg, J., Harati, A., & Picone, J. (2013). A Comparative Analysis of Bayesian Nonparametric Inference Algorithms for Acoustic Modeling in Speech Recognition. *Proceedings of International Joint Conferences on Computer, Information, Systems Sciences, & Engineering* (pp. 1–5). Bridgeport, Connecticut, USA: Springer-Verlag.
20. Harati, A., Choi, S. I., Tabrizi, M., Obeid, I., Jacobson, M., & Picone, J. (2013). The Temple University Hospital EEG Corpus. *Proceedings of the IEEE Global Conference on Signal and Information Processing*. Austin, Texas, USA.
21. Harati, A., & Picone, J. (2013). Speech Acoustic Unit Segmentation Using Hierarchical Dirichlet Processes. *Proceedings of INTERSPEECH*. Lyon, France.
22. Harati, A., & Picone, J. (2013). Assessing Search Term Strength in Spoken Term Detection. *Proceedings of the IEEE International Multi-Disciplinary Conference on Cognitive Methods in Situation Awareness and Decision Support* (pp. 114 – 117). San Diego, California, USA.
23. Harati, A., Picone, J., & Sobel, M. (2012). Applications of Dirichlet Process Mixtures to Speaker Adaptation. *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing* (pp. 4321–4324). Kyoto, Japan.
24. A. Harati, M. Ahmadian and A. Mosavinia, “A New Criterion To Compare Different LDPC Codes For Hardware Implementation,” *Proceedings of the 21st IEEE Canadian Conference on Electrical and Computer Engineering* (CCECE 2008), Niagara Falls,

PUBLICATIONS:
Conference
Presentations

Canada, pp. 805-808, May 2008.

25. Harati, A., Golmohammadi, M., Jacobson, M., Lopez, S., Obeid, I., Picone, J., & Tobochnik, S. (2016). Automatic Interpretation of EEGs for Clinical Decision Support. In Proceedings of the American Clinical Neurophysiology Society (ACNS) Annual Meeting (p. 1). Orlando, Florida, USA.
26. Obeid, I., Harati, A., & Picone, J. (2014). EEG Event Detection Using Big Data. *Proceedings of the 48th Annual Asilomar Conference on Signals, Systems, and Computers* (p. TBD). Pacific Grove, California, USA.
27. Adamczyk, M., Parker, D., Cremer, J., Picone, J., Harati, A., & Polgreen, P. (2013). A mobile phone application for pertussis classification based on machine learning. *Proceedings of the International Meeting on Emerging Diseases and Surveillance*. Vienna, Austria.

PUBLICATIONS:
Book Chapters, PhD
Dissertation,
MS Thesis,
BSC Project

28. Harati, A., Rutowski, T., Lu, Y., Chlebek, P., Oliveira, R., Shriberg, E., & Lin, D. (2022). Generalization of Deep Acoustic and NLP Models for Large-Scale Depression Screening. In *Biomedical Sensing and Analysis: Signal Processing in Medicine and Biology* (pp. 99-132). Cham: Springer International Publishing.
29. Harati, A.. (2015). *Nonparametric Bayesian Approaches for Acoustic Modeling*. PhD Dissertation, Department of Electrical and Computer Engineering, Temple University, Philadelphia, Pennsylvania, USA, May 2015.
30. Harati, A.. (2007). *Simulation and DSP Implementation of an LDPC Codec to determine the effective parameters for its performance*. Master Thesis, K. N. Toosi University of Technology, Tehran, Iran, December 2007.
31. Harati, A. (2004). Study and Simulation of Digital Radio Mondiale (DRM) Systems. B.S.C Final Project, Tabriz University, Tabriz, Iran, October 2004.

**WORK
EXPERIENCE**

Jan 2019 – Present : Machine Learning Engineer & Founder | ideaLabs Inc.

- **Technical Consultancy:** Provide end-to-end machine learning solutions, specializing in NLU, NLP, acoustic modeling, and risk modeling. Partnered with healthcare and enterprise clients to architect and deploy scalable production-grade AI systems.
- **Strategic Development:** Led long-term machine learning initiatives for firms such as Ellipsis Health, transforming raw speech and behavioral data into actionable clinical insights.
- **Product Innovation & R&D:** Drive the full lifecycle of proprietary consumer and B2B AI products. Maintain a private R&D pipeline focused on state-of-the-art developments in generative AI and signal processing.
- **Software Development & Open Source:** Maintain and contribute to multiple open-source and closed-source repositories focused on advancing machine learning tools, building agentic AI systems, and developing robust software. Projects and documentation are available across both [GitHub](#) and [ideaLabs.ai](#).

Jan 2019 – Aug 2024 : Machine Learning Consultant | Ellipsis Health Inc.

- Architected and delivered specialized machine learning solutions for behavioral health, focusing on speech and acoustic modality integration to support clinical assessment tools.

May 2020 – Present : Lead Machine Learning Engineer & Co-Founder | SusuRobo Inc.

- Lead the development of generative AI solutions designed to enhance interactivity and engagement for streamers and content creators.

May 2018 – Jan 2019 : Machine Learning and Speech Recognition Engineer | Ellipsis Health Inc.

- Engineered algorithms and models for detecting depression from speech acoustic and linguistic content; successfully deployed production-grade pipelines for clinical applications.

July 2015 – April 2018 : Software Developer and Speech Recognition Engineer | Jibo Inc.

- Developed core speech recognition software and machine learning algorithms for social robotics, advancing capabilities for human-robot interaction.

Oct 2015 – Present : Scientific Adviser & Co-Founder | BioSignal Analytics

- Provide scientific leadership and strategic direction for the development of advanced biosignal processing and analytics technologies.

Jan 2014 – July 2015 : Graduate Research Assistant & Lead Researcher | Temple University

- Developed novel machine learning algorithms for the automatic interpretation of clinical EEG signals collected from Temple University Hospital.

Sept 2012 – Dec 2013 : Teaching Assistant | Temple University

- Facilitated instruction for courses in communication systems and embedded systems within the Department of Electrical and Computer Engineering.

Jan 2010 – Aug 2012 : Graduate Research Assistant | Temple University

- Conducted advanced research in acoustic modeling for speech recognition; developed innovative machine learning approaches for large-vocabulary continuous speech recognition.

Feb 2007 – Dec 2009 : Signal Processing Researcher | Bina-Pardaz Company

- Conducted research and development in signal processing applications.

Sept 2007 – Jan 2008 : Graduate Research Assistant | K.N. Toosi University of Technology

- Supported research initiatives in electrical engineering and signal processing.

**PROFESSIONAL
MEMBERSHIPS**

Senior Member, IEEE, 2016 – Present
Member, Eta Kappa Nu (HKN), 2011 – Present

REFERENCES

References will be provided upon request.